

MONITORING TRAINING LOADS IN PROFESSIONAL BASKETBALL PLAYERS ENGAGED IN A PERIODIZED TRAINING PROGRAM

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ABSTRACT

Aoki, MS, Ronda, LT, Marcelino, PR, Drago, G, Carling, C, Bradley, PS, and Moreira, A. Monitoring training loads in professional basketball players engaged in a periodized training program. *J Strength Cond Res* 31(2): 348–358, 2017—The aim of this study was to investigate the dynamics of external training load (eTL) and internal training load (iTL) during seasonal periods, and examine the effect of a periodized training program on physical performance in professional basketball players. Repeated measures for 9 players (28 ± 6 years; 199 ± 8 cm; 101 ± 12 kg) were collected from 45 training sessions, over a 6-week preseason phase and a 5-week in-season phase. Physical tests were conducted at baseline (T1), week 4 (T2), and week 9 (T3). Differences in means are presented as $\% \pm$ confident limits. A very likely difference was observed during in-season compared with preseason for the eTL variables (measured by multivariable monitoring device), mechanical load (13.5 ± 8.8) and peak acceleration (11.0 ± 11.2), respectively. Regarding iTL responses, a very large decrement in TRIMP (most likely difference, -20.6 ± 3.8) and in session rating of perceived exertion training load (very likely difference, -14.2 ± 9.0) was detected from preseason to in-season. Physical performance improved from T1 to T3 for Yo-Yo intermittent recovery test 1 (62.2 ± 34.3 , effect size [ES] > 1.2); countermovement jump (8.8 ± 6.1 , ES > 0.6); and squat jump (14.8 ± 10.2 , ES > 0.8). Heart rate (HR; %HRpeak) exercise responses during a submaximal running test decreased from T1 to T3 (3.2 ± 4.3 , ES < 0.6), as well as the HR recovery after the test (14.7 ± 8.8 , ES > 1.2). These

results provide valuable information to coaches about training loads and physical performance across different seasonal periods. The data demonstrate that both eTL and iTL measures should be monitored in association with physical tests, to provide a comprehensive understanding of the training process.

KEY WORDS training load, periodization, performance, training monitoring

INTRODUCTION

A major challenge in sport is to appropriately prescribe training loads (TLs) to maximize athletes' physical performance while avoiding overtraining and injury, during the competitive season (8,22,45). A general consensus is that appropriate periodization is a key factor in achieving optimal performance outcomes (23,40,45). However, limited empirical evidence supports this statement in team sports (36).

Basketball is an intermittent team sport which demands a wide range of physical requirements such as repeated sprint ability (RSA), and changes in running direction and speed, jumps, and high-intensity running (4,14). To maintain performance in these physical requirements, appropriate assessment and adequate periodization are essential. Akin to other team sports, monitoring TL in basketball is especially important, because, within team members, different responses to training sessions and matches occur. Given that basketball-specific actions involve whole-body displacement in forward, backward, lateral, and vertical directions, accelerometer-derived measures are therefore, a useful approach to monitor external training load (eTL) in basketball players (42). However, there has been a paucity of research in examining eTL in conjunction with internal training load (iTL) in basketball (42).

The monitoring of eTL measures derived from triaxial accelerometers is now considered a viable tool in team sports (1,6,35). For instance, Boyd et al. (6) described the

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eTL of Australian football performed in official matches and training sessions, using a triaxial accelerometer. More recently, Arruda et al. (1) suggested that accelerometer-derived measures might be used as an alternative method to assess eTL in soccer. Accelerometers have also been used for determining the demands of practice and competition in basketball players (35). Although advancing knowledge of periodized eTL in team sports is of great importance, it is also imperative to recognize that eTL only describes the activity that a player has completed, and may not accurately depict the physiological stress imposed on individual athletes (30). Accordingly, iTL has been used to monitor an athlete's response to a training dose (13,24,30–31,34,38,41). Taking into account the challenges in measuring the various types of stresses encountered during training, the session rating of perceived exertion (session-RPE) method has been largely adopted (13,24,30–31,34,38,41) to assess iTL. The session-RPE method has been validated for use in a number of team sports, demonstrating strong correlation with both internal physiological responses (22,38,44) and external measures of TL (17–18,22,24,30). As such, further investigation is warranted to advance current knowledge with regard to the association between eTL through accelerometer-derived measures, and iTL, during specific-basketball training sessions, across both preseason and in-season periods. This information may be used to design effective training strategies that may improve performance and reduce injury risk. Therefore, this study aimed to (a) examine the effect of a periodized training plan on physical performance measures and (b) describe TL (i.e., eTL using accelerometer-derived measures, and iTL through heart rate [HR] and session-RPE measures) associated with technical and tactical sessions, in professional basketball players during preseason and in-season periods.

METHODS

Experimental Approach to the Problem

The study was conducted during the 2014–2015 competitive Brazilian professional basketball season. Data were collected from 45 basketball-specific team-training sessions (i.e., technical and tactical sessions), performed across an 11-week period. The study period was divided into 2 phases: preseason phase (6-week period) and in-season phase (5-week period). The preseason phase was divided into preseason 1 and preseason 2, with respect to the main technical and tactical training contents for each period. During the preseason 1 period (week 1–3), 4 technical and tactical training sessions were performed per week, with a total duration ranging from 60 to 70 minutes per session. During the preseason 2 period (week 4–6), 5 to 6 technical and tactical sessions were performed per week (duration range 80–100 minutes each). No matches were played during the preseason. During the in-season period, session durations ranged from 70 to 80 minutes. Four technical and tactical sessions were performed per week, and the team competed in 2 games per week (i.e., Wednesday and Saturday). All on-

court team-training sessions performed during the study period were considered for this study. The physical conditioning sessions are described to illustrate the whole training periodization approach. Table 1 details the training content for each of the phases, taking into account the technical and tactical and physical sessions. Table 2 shows the typical daily physical training exercises during preseason (first and second period) and in-season phases represented by week 2, week 5, and week 8, respectively. Physical performance tests were conducted at the beginning of the study (T1), after week 4 (T2), and after week 9 (T3).

Subjects

A convenience sample of 9 professional male basketball players (27.8 ± 6.4 years; stature: 199.1 ± 8.3 cm; body mass: 101.3 ± 12.1 kg) from an initial sample of 14 players were recruited from a professional team competing in the main State Basketball Championship and the National Brazilian League. The 5 players who did not participate in all training sessions during the study period were not included in the analysis. The assessed players were classified into 3 playing positions: point-guards = 2; wings = 4; centers = 3. During the assessed in-season phase (week 7–11), the team participated in 10 official matches (5 away matches and 5 home matches). The team ended this investigated phase with 8 wins and 2 defeats. All 10 matches were played during the regular season (qualifying phase to playoffs), and the team was qualified in first place, therefore, advancing to the playoffs. All players and coaches were informed about the research protocol in terms of requirements, benefits, and risks. Their written consent was obtained before the study began. There were no players under the age of 18 years. This research conforms to the ethical principles, which was approved by the local university research ethics committee.

Procedures

Technical and Tactical Training Sessions. All training sessions were performed on the same regulation court under controlled environmental conditions. The usual verbal encouragement from the head coach and staff members was permitted during sessions. All training sessions started with a 15-minute standardized warm-up based on running, technical skills (ball dribbling and layups), full-court offense drills (e.g., 3 vs. 0; 3 vs. 2; and 3 vs. 3), and dynamic stretching exercises. During the first period of the preseason (weeks 1–3), the sessions were organized as follows: tactical drills without opposition (2 vs. 0 to 5 vs. 0) focusing on offensive aspects; tactical drills with opposition (1 vs. 1 to 4 vs. 4) focusing on defensive aspects and technical drills (e.g., shooting, passing). During the second period of the preseason (weeks 4–6), the main content of the sessions was tactical drills without the opposition (3 vs. 0 to 5 vs. 0), half-court scrimmage (4 vs. 4 and 5 vs. 5), and full-court scrimmage (4 vs. 4 and 5 vs. 5). During the in-season, the technical and tactical training sessions were very similar to those performed during the second period of the preseason,

however, the intensity and volume of the training sessions were constantly manipulated using iTL and eTL responses. The manipulation of the intensity included changes in the work to recovery ratio within and between exercises, varying the number of players performing full-court scrimmage exercises, changing rules (i.e., size of court, number of players, playing with or without free shots, and inclusion of repeated sprints after a given playing situation). For all training sessions (i.e., preseason and in-season phases), players were allowed to consume water ad libitum during recovery periods.

Physical Training Sessions. All physical training sessions started with a specific warm-up (i.e., light weightlifting exercises, and jumps). During the first period of preseason (weeks 1–3), 6 physical training sessions per week were performed with a mean duration of ~90 minutes; the main goal of this period was to develop athletes' strength; a secondary goal was intended to build speed and agility and endurance capacities. During the second period of preseason (weeks 4–6), 5 physical training sessions were performed per week,

with a mean duration of ~65 minutes. During this period, the main goal was to develop the players' power, speed and agility; a secondary goal was to maintain the previous strength level achieved during the first period of preseason (week 1–3). During the in-season (weeks 7–11), 3 sessions per week were performed with a mean duration of ~60 minutes. The main goal of this phase was to provide a sufficient stimulus to maintain the players' previously achieved level of fitness, while providing recovery from technical and tactical sessions and matches.

Training Load Measures. The players' physical activity during each technical and tactical training session was monitored using a multivariable monitoring portable device (Bioharness; Zephyr Technology, Auckland, New Zealand). Devices were placed on the middle of a players' chest. The players used the same unit across all training sessions.

Internal Training Load Measures. **Heart Rate.** Data were captured using a chest strap and reported as $\text{b} \cdot \text{min}^{-1}$, and

TABLE 1. Typical daily training content during preseason and in-season phases.

	Preseason		In-season
	Week 2	Week 5	Week 8
Monday	Morning: physical training, strength training, and power (jumps) and speed Afternoon: technical tactical session	Morning: physical training, strength and power training—(complex training) Afternoon: technical and tactical session	Morning: physical training, strength and power training Afternoon: technical and tactical session
Tuesday	Morning: physical training, strength training	Morning: physical training, speed and agility + core training Afternoon: technical and tactical session	Morning: physical training, speed and agility + core training Afternoon: technical and tactical session
Wednesday	Morning: physical training, strength training, endurance (interval training)	Morning: physical training, strength and power training—complex training Afternoon: technical and tactical session	Afternoon: technical and tactical session
Thursday	Morning: physical training, strength training Afternoon: technical tactical session	Morning: physical training, speed and agility + core training Afternoon: technical and tactical session	Evening: game
Friday	Morning: physical training, strength training power (jumps) and speed Afternoon: technical and tactical session	Morning: physical training, strength and power training—(complex training) Afternoon: technical and tactical session	Morning: technical and tactical session
Saturday	Morning: physical training, strength training, endurance (interval training)	Rest	Evening: game
Sunday	Off	Off	Off

Banister's TRIMP (training impulse) values were also calculated (see Formula):

$$t \times \text{HR}_R \times e^{1.92 \times \text{HR}_R}$$

where, t is training session duration in minutes, e is base of the natural logarithm (2.712), and reserve HR (HR_R) is determined by the following equation:

$$\text{HR}_R = \frac{\text{HR}_S - \text{HR}_B}{\text{HR}_{\text{max}} - \text{HR}_B}$$

where, HR_S is session HR, HR_B is rest HR, HR_{max} is maximum HR.

The HR_{max} values were obtained before data collection by performing the Yo-Yo intermittent recovery test level 1 (Yo-Yo IRT1), as previously done in basketball players (34,41).

Physiological Load. This is a HR-based index, a value derived from the sum of all obtained physiological values (physiological intensity) from the training session, and the time spent at a given intensity zone. The intensity zone is determined based on the subject's maximum heart rate (% of HR_{max}). The intensity level was classified from 0 to 10 as follows: 0 $\leq$ 50% HR_{max} ; 1 > 50–55% HR_{max} ; 2 > 55–60% HR_{max} ; 3 > 60–65% HR_{max} ; 4 > 65–70% HR_{max} ; 5 > 70–75% HR_{max} ; 6 > 75–80% HR_{max} ; 7 > 80–85% HR_{max} ; 8 > 85–90% HR_{max} ; 10 > 95 to 100 or >100% HR_{max} . This zone classification system forms the basis of the analysis performed by the software (Bioharness; Zephyr Technology), and therefore adopted in this study in accordance with manufacturer's guidelines.

Session Rating of Perceived Exertion. Players' session-RPE were recorded ~30 minutes after each session using the Borg

TABLE 2. Typical daily physical training exercises during preseason (first and second preseason periods) and in-season phases represented by week 2, week 5, and week 8, respectively.*

Week 2	
Monday, Wednesday, Friday	Strength training (4–5 sets $\times$ 12 RM) Hamstring leg curl, hip adductor, hip abductor, forehead triceps, pulley triceps, dumbbell biceps curl, w bar biceps curl Power (4 sets $\times$ 4 reps; maximal velocity) Hurdle jump, hurdle jump with fright, hurdle one-step, jump, barrier lateral jump Speed (6–8 sets $\times$ 10–15 meters; running with speed variation) Endurance (interval training)—interval training 8–10 $\times$ 2' running: 30" rest†
Tuesday, Thursday, Saturday	Strength training (4–5 sets $\times$ 12 RM) Deeps squats 2–4 $\times$ 12, leg press, leg curl, dead lift machine, bench press, pull-up Endurance (interval training)‡
Week 5	
Monday, Wednesday, Friday	Strength and power training—(complex training) Split squat 2–3 $\times$ 8, weightlifting: clean 3–4 $\times$ 6 (power), leg press 3–4 $\times$ 6 RM + hurdle jump 3–4 $\times$ 2, dead lift 3–4 $\times$ 6 RM + hurdle jumps 3–4 $\times$ 2, bench press 3–4 $\times$ 6 RM + chest medicine ball throwing 3–4 $\times$ 4 Jumping push-ups 3–4 $\times$ 6
Tuesday, Thursday, Saturday	Speed and agility Sprint 6–8 $\times$ 15–20 m, sprint with 1–2 changes of direction 8–10 $\times$ 20–25 m Core training: elbow plank 3 $\times$ 60", side plank 3 $\times$ 60", sit-up 4 $\times$ 15
Week 8	
Monday, Wednesday, Friday	Strength and power training Split squat 2 $\times$ 8 RM, weightlifting: clean 3 $\times$ 4 (power), leg press 3–4 $\times$ 6 RM Dead lift 3–4 $\times$ 6 RM, bench press 3–4 $\times$ 6 RM, hurdle jump 3–4 $\times$ 2, box jumps 3–4 $\times$ 2, medicine ball chest throwing 3–4 $\times$ 4, jumping push-ups 3–4 $\times$ 6
Tuesday, Thursday, Saturday	Speed and agility Footwork ladder agility 4–5 $\times$, sprint 4–5 $\times$ 10 m sprint, change of direction drills 4–5 $\times$ 15 m, core training: full plank 3 $\times$ 60", side rotation plank 3 $\times$ 60" V-seat with trunk rotation 4 $\times$ 15

*RM = repetition maximum.

†Only on Wednesday.

‡Only on Saturday.

10-point scale (CR-10 scale) (18). Session-RPE was derived by asking each player “How intense was your session?” Players were already familiarized with the CR-10 scale before this study. Daily TL was calculated by multiplying session-RPE by the session duration (18). The validity of using session-RPE for monitoring training and competition loads in basketball players has previously been demonstrated (1,31,37–38).

External Training Load Measures. To monitor *mechanical load (ML) and acceleration*, players used the multivariable monitoring portable device (Bioharness; Zephyr Technology). The triaxial accelerometer (piezoelectric technology) is a micro electromechanical sensor. It is sensitive along 3 orthogonal axes (vertical [x], sagittal [z], and lateral [y]). Acceleration data were measured as gravitational forces (*g*) (−3 to +3*g*) on each single axis or as vector magnitude units (VMUs), which is an averaged value of the previous 1-s epoch (25–27). The reliability and validity of the Bioharness has been demonstrated in both laboratory and field-based environments (25–27). Mechanical load is the accumulation of the mechanical intensity value over time. The ML was measured using VMU, $\sqrt{(x^2+y^2+z^2)}$; (vertical [x], sagittal [z] and lateral [y]). The time spent in each mechanical zone was considered for analysis. Intensity is defined according to

the range in which the peak (in any axis) acceleration *g* value fits in any 1-s epoch. Value scaled linearly between 0.5 (0) and 3.0 *g* (10) as follows: 0 = 0.50; 1 = 0.75; 2 = 1.00; 3 = 1.25; 4 = 1.50; 5 = 1.75; 6 = 2.00; 7 = 2.25; 8 = 2.50; 9 = 2.75; and 10 = 3.00 VMU.

Physical Assessments. Submaximal Running Test. A submaximal 5-minute running and 5-minute recovery test (7) was performed at the beginning of every testing session. All players were tested together with the intensity of the exercise bout fixed at 10 km · h^{−1} across 40-m shuttles. Heart rate during exercise and during recovery were assessed as previously described, using a harness (Bioharness; Zephyr Technology). Buchheit et al. (8) demonstrated that HR during exercise is a useful variable for monitoring positive training responses during training, and reported a 3.3% coefficient of variation (CV) for this measure, largely affected by daily variations in training load, suggesting that this measure is a valid and highly reproducible measure.

Yo-Yo Intermittent Recovery Level 1 Test. The Yo-Yo IRT1 test was applied according to previously described methods to assess the players’ aerobic and anaerobic capacity (28,34). All players were familiar with the testing procedures. The players performed repeated 20-m

TABLE 3. Descriptive analysis for physical performance.*†

Test	Variable	Baseline (T1)	Week 4 (T2)	Week 9 (T3)	Difference in means (%) ± 90% CL
5'–5' submaximal run	HRE (%) HRpeak	87.1 ± 6.8	83.6 ± 5.8	82.4 ± 5.5	(A) −3.9 ± 3.8‡ (B) −3.2 ± 4.3§ (C) 0.8 ± 1.3
	HRR (%) HRpeak	25 ± 1.9	27.4 ± 4.2	29.6 ± 2.8	(A) 12.3 ± 18.7 (B) 14.7 ± 8.8 (C) 8.1 ± 10.5‡
Yo-Yo IRT1	Distance (m)	1,120 ± 412.5	1,355 ± 466.2	1,737.1 ± 515.3	(A) 31.8 ± 14.3 (B) 62.2 ± 34.3 (C) 36.0 ± 39.2‡
Squat jump	Height (cm)	34.8 ± 2.6	37.8 ± 2.3	40.7 ± 3.6	(A) 9.5 ± 5.7 (B) 14.8 ± 10.2 (C) 7.4 ± 4.1
Countermovement jump	Height (cm)	38.1 ± 2.8	39.4 ± 3.5	41.9 ± 4.1	(A) 4.3 ± 4.1‡ (B) 8.8 ± 6.1 (C) 5.4 ± 3.8‡
Repeated sprint Test	Best time (s)	3.20 ± 0.14	3.11 ± 0.12	¶	(A) −2.7 ± 1.5 (A) −1.9 ± 2.3§
	Mean time (s)	3.33 ± 0.16	3.26 ± 0.16	¶	

*CL = confidence limit; HRE = heart rate exercise; HRR = heart rate rest; Yo-Yo IRT1 = Yo-Yo intermittent recovery test level 1.

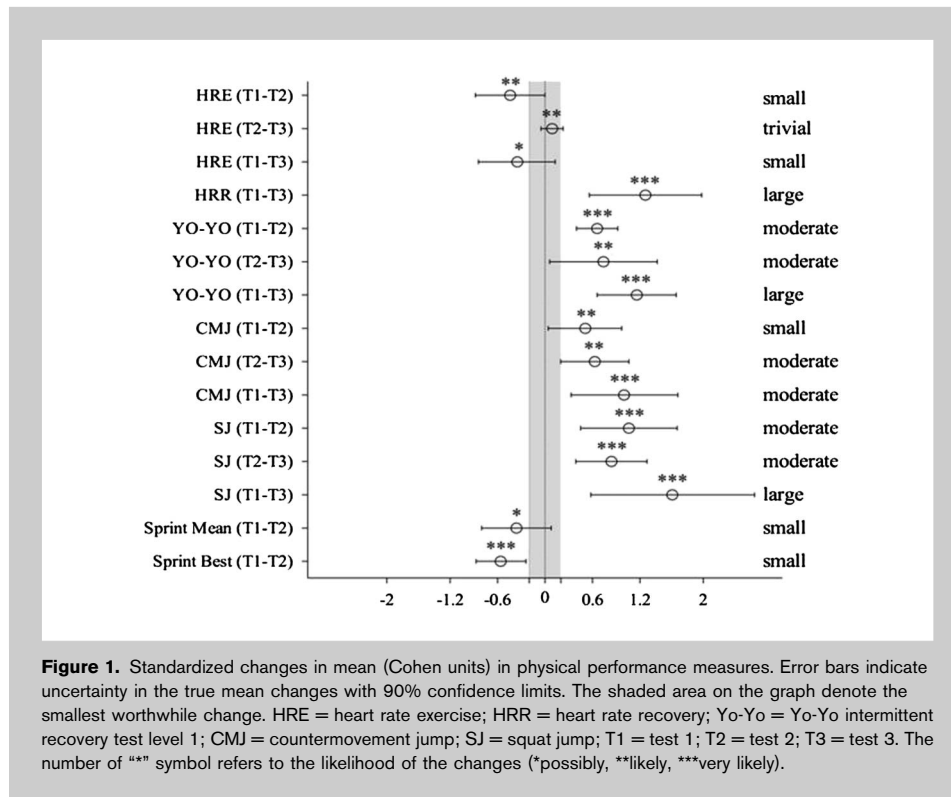
†Differences in means (%) ± 90% CL are identified as (A) baseline-week 4; (B) baseline-week 9; (C) week 4–week 9.

‡Likelihood for the magnitude of the true effect = likely.

§Likelihood for the magnitude of the true effect = possible.

||Likelihood for the magnitude of the true effect = very likely.

¶The RST was not conducted due to operational problems.



shuttles, back and forth between the starting line and finish line marked by cones, at progressively increasing speeds dictated by an audio bleep emitting from a compact disc player. Between each shuttle, the players had a 10-second period of walking around a cone placed 5 m from the starting line. Failure to achieve the shuttle run on 2 successive occasions resulted in termination of the test. Total distance covered represented the test result. All tests were performed on the same basketball court where the players

trained. The reliability of this test has been previously quantified with a CV of ~5% (28).

Vertical Jumps. Players performed both countermovement jump (CMJ) and squat jump (SJ). The CMJ was initiated from a standing position to a self-selected depth position, and performed as quickly as possible with maximal effort. The hands were placed on the hips during the entire CMJ movement. Although no restrictions were placed on the knee angle attained during the eccentric phase of the jump, players were instructed to maintain straight legs during the flight. The SJ was performed as a concentric only movement (i.e., no countermovement before takeoff) with hand placement on the hips to remove the effects of arm swing. The initial squat jump angle position was required to be at 90°. An adjustable bench was used as reference to keep the required SJ position angle. Three jumps were performed for each type of jump with a 2-minute rest period between jumps. All jumps were conducted on an electronic jump mat (Ergojump Jump Pro 2.0; CEFISE, Nova Odessa, São Paulo, Brazil). Jump mat technology provides valid measures of jump height compared with a criterion system ($r = 0.97$) (29), and pilot testing for the current study indicated that the jump mat system also

TABLE 4. Descriptive analysis for internal and external load variables and differences between training phases (preseason and in-season).*

Variable	Preseason	In-season	Difference in means (%) $\pm$ 90% CL	Uncertainty in the true differences
Physiological load (AU)	242.5 $\pm$ 55.9	230.6 $\pm$ 34.6	-3.4 $\pm$ 11.0	Unclear
HRavg ($b \cdot min^{-1}$)	115.8 $\pm$ 7.8	120.2 $\pm$ 7.1	3.9 $\pm$ 2.2	Very likely
s-RPE (AU)	4.8 $\pm$ 1.0	5.2 $\pm$ 0.9	8.3 $\pm$ 9.3	Likely
TRIMP	27.1 $\pm$ 2.1	21.5 $\pm$ 1.6	-20.6 $\pm$ 3.8	Most likely
TL (AU)	442.9 $\pm$ 89.2	377.1 $\pm$ 68.3	-14.2 $\pm$ 9	Very likely
Mechanical load (AU)	172.6 $\pm$ 26.3	195.2 $\pm$ 23.6	13.5 $\pm$ 8.8	Very likely
Peak acceleration ($m \cdot s^{-2}$)	2.2 $\pm$ 0.2	2.4 $\pm$ 0.2	11.0 $\pm$ 11.2	Very likely
Volume (min)	91.4 $\pm$ 2.8	70.5 $\pm$ 1.7	-22.8 $\pm$ 1.8	Most likely

*CL = confidence limit; s-RPE = session rating of perceived exertion; TL = training load (s-RPE multiplied by training session duration); AU = arbitrary units.

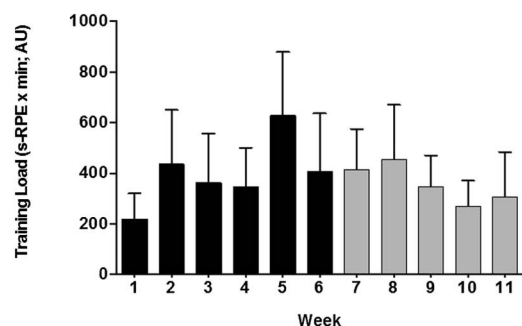


Figure 2. Training load data (session-RPE $\times$ minute; AU) across the 11-week microcycle during both preseason (weeks 1–6) and in-season (weeks 7–11) phases (technical and tactical training sessions). RPE = rating of perceived exertion; AU = arbitrary units.

provides reliable measures ($CV < 5.0\%$; Intraclass Correlation Coefficient [ICC] > 0.90) for both SJ and CMJ).

Repeated Sprints Ability Test. The protocol adopted in this study was 12×20 m sprints with 20 seconds of active rest between sprints as used by Meckel et al. (33). A photoelectric cell timing system (Multisprint; Hidrofit, Minas Gerais, Brazil) was used to record the duration of each sprint (accuracy of 0.001 seconds). Two sets of timing gates were used, one for the start (opening gate) and one for the end (closing gate). Players were instructed to decelerate only after the closing gate, and return to the start point to prepare for the next sprint. Although the return pace was chosen by each player, the instructor provided verbal feedback about the remaining recovery time. A standing start with the front foot placed 50 cm behind the opening gate was used for all sprints. All players received verbal encouragement during the test from instructors and coaches. The best sprint time (BT) and the mean sprint time (MT) were registered. The test-retest reliability previously reported for RSA total running time is 0.94 (16).

Statistical Analyses

Descriptive values are shown for technical and tactical, and physical training loads performed over the 11 training weeks. A magnitude-based inferential statistical approach was adopted based on previous recommendations (2,47). Param-

eters were log-transformed to reduce bias due to the non-uniformity of error, and analyzed using a customized Excel spreadsheet (19–20). The effect size (ES) was calculated to determine the meaningfulness of the difference between pre-season and in-season phases, corrected for bias using Hedges formula and presented with 90% confidence limits (CL) (2,12). The ES magnitudes were classified as trivial (<0.2), small (>0.2 – 0.6), moderate (>0.6 – 1.2), and large (>1.2). Smallest worthwhile differences were estimated from the standardized units multiplied by 0.2. Uncertainty in the true differences of the scenarios was assessed using nonclinical magnitude-based inferences (21). Values are presented as means and standard deviations unless otherwise stated.

RESULTS

Physical Test Performances

Physical performance measures over the competitive season are presented in Table 3 and Figure 1. Large improvements in Yo-Yo IRT1 performance (difference in means; $\% \pm CL = 62.2 \pm 34.3$; $ES > 1.2$) and a moderate-to-large increase in CMJ (8.8 ± 6.1 ; $ES > 0.6$) and SJ (14.8 ± 10.2 ; $ES > 0.8$) performances were found from baseline (T1) and after a 9-week period (T3). Small-to-moderate improvements in RSA test were observed from T1 to T2 (-2.7 ± 1.5 , and -1.9 ± 2.3 ; $ES < 0.6$; for MT and BT, respectively). Heart rate exercise (HRE) response during the running test decreased from T1 to T3 (3.2 ± 4.3 ; $ES < 0.6$) as well as the heart rate recovery (HRR) (14.7 ± 8.8 ; $ES > 1.2$).

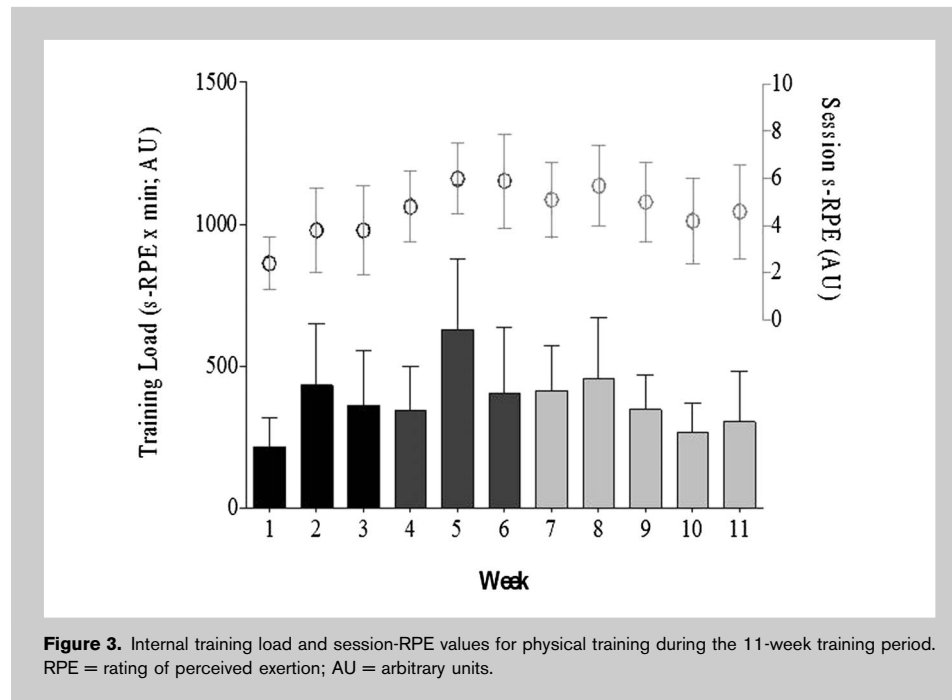
Overview of Weekly External Training Load and Internal Training Load

Descriptive values for technical and tactical iTL and eTL, and differences between preseason and in-season, are presented in Table 4. Mean training volume (min) between periods showed very large differences, with lower values during the in-season phase (-22.8 ± 1.8). Regarding iTL responses, this reduction (from preseason to in-season) was accompanied by a very large decrement in TRIMP (most likely, -20.6 ± 3.8) and in session-RPE TL (very likely, -14.2 ± 9.0). However, the effect of such reduction was only trivial (unclear reduction; $-3.4\% \pm 11.0$) for the physiological load measure. On the other hand, HR average and session-RPE showed increases from preseason to in-season

TABLE 5. iTL–physical training; descriptive values expressed as average per week.*

Variable	Preseason first period	Preseason second period	Preseason	In-season
Accumulated iTL (AU)	$3,591.3 \pm 877.6$	$1,724.8 \pm 347.7$	$2,621.2 \pm 545.1$	754 ± 110.3
Average iTL (AU)	638.7 ± 127.3	357.9 ± 56.9	489.4 ± 85.6	251.3 ± 29.1
s-RPE (AU)	6.3 ± 1.1	5.2 ± 0.9	5.7 ± 0.9	4.8 ± 0.7

*iTL = internal training load; s-RPE = session rating of perceived exertion; AU = arbitrary units.



with a very likely and likely effect (3.9 ± 2.2 , and 8.3 ± 9.3 , respectively). Moreover, the ML and peak acceleration (eTL measures) showed very likely differences with higher values for in-season phases (13.5 ± 8.8 , and 11.0 ± 11.2 , respectively). The Figure 2 presents the TL performed over the 11-week microcycles during both preseason (week 1–6) and in-season (week 7–11) phases, for technical and tactical training sessions. Regarding the physical training, the descriptive values are presented in Table 5, and the iTL and session-RPE scores for physical training during the 11-week training period are displayed in Figure 3.

DISCUSSION

This study aimed to (a) examine the effect of a periodized training plan on physical performance test measures in professional basketball players during preseason and in-season phases and (b) describe eTL using accelerometer-derived measures and the iTL associated with technical and tactical sessions. The main findings observed were a large improvement in Yo-Yo IRT1 performance and RSA, a moderate-to-large increase in jumping performance, and a decreased submaximal HR response to a 5-minute submaximal run test. Moreover, training volume was higher during the preseason phase whereas ML and peak acceleration values were greater during the in-season phase.

These findings have practical implications for coaches concerning the distribution of training content and strategies during preseason and in-season phases in basketball. Although the improvement in Yo-Yo IRT1, RSA, and jump performances are associated with general training-induced

adaptations, the improvements observed during the in-season phase suggest that these indicators might also be viewed as viable indicators of basketball-related physical performance measures, and that they might relate to competitive performance. It is noteworthy that the improvements in this study are in accordance with previous findings (41). These authors assessed 19 elite female basketball players who were members of the Brazilian national team, preparing for the FIBA Americas Championship. The experimental design included 1 overloading period of 3-weeks duration (weeks 4–6), followed by a 1-week tapering phase (week 7). The second overloading period was 3 weeks in duration (week

8–10), followed by a 2-week tapering phase (weeks 11–12). Strength, jumping power, endurance, and agility were assessed, and iTL and recovery stress-state were also quantified weekly. The results of this strategy revealed an improvement in strength and jumping performances from pre- to post-training. The speed, agility, and intermittent endurance capacity also improved post-training.

It is noteworthy that the improvements observed in this study, notably for Yo-Yo IRT1 suggests the efficacy of the adopted present training periodization strategy. In this study, the observed difference in means ($\% \pm \text{CL}$) for Yo-Yo IRT 1 from the beginning of the study to the ninth week was 62.2 ± 34.3 , with the magnitude of the effect being classified as very likely. This effect is highly relevant taking account that it was observed for professional players with a 9-week training investigation. For example, the improvement is higher than that reported by Nunes et al. (41) after 12 training weeks ($\text{ES} = 0.35$), or in other team sport athletes such as those reported by Fanchini et al. (15) with 11 soccer players whose Yo-Yo IRT1 performance increased 40% after training (90% confidence interval = 30.7–50.4%) within a 17-week period investigation; the present improvement in Yo-Yo IRT1 is also higher than that reported by Buchheit et al. (10) who reported that Yo-Yo IRT1 running performance in temperate conditions (22°C) was improved on average by 7% after a training week in 15 soccer players belonging to a Faroe Island first division team and a Danish second division team. The present change in performance therefore, may be interpreted as remarkable, especially taking into account that it was achieved in already well-trained professional basketball players.

The results derived from the physical performance tests employed in this study suggested that the tests are highly sensitive to training-induced improvements in physical capacity indices. Moreover, it is worth mentioning that the observed improvement in RSA, jumping, and running performance need to be contextualized. Basketball is an intermittent activity characterized by frequent, high-intensity actions and change of directions across short distances (3,4,32). Basketball also incorporates sprints, movements involving accelerations and decelerations, and jumps (3,4,14,32). With this in mind, it seems reasonable to assume that the physical tests used in this study can confidently be adopted in a way within an applied setting to systematically aid in planning a training.

Together with the physical performance tests, there was a decrease in the players HR responses to the 5-minute submaximal run from T1 to T3. These noninvasive and nonexhaustive measures of assessing submaximal aerobic capabilities, despite some reported limitations (9) are considered an index of cardiorespiratory fitness, which is strongly correlated with running performance (7,9). In a study with Australian football players, Buchheit et al. (8) demonstrated the usefulness of submaximal HR responses as a monitoring tool during a preseason period. Buchheit et al. (8) adopted a submaximal 5-minute run and 5-minute recovery test (7) to assess the training status of the players. This test was performed at the start of every training session. The intensity of the exercise bout was fixed at $13 \text{ km} \cdot \text{h}^{-1}$ across 40-m shuttles, and HR response was assessed as previously described (7). In this study, however, the 5-5 submaximal run was performed at the start of the session with the intensity of the exercise bout fixed at $10 \text{ km} \cdot \text{h}^{-1}$ across 40-m shuttles. This adaptation in the intensity of the exercise was necessary to maintain the submaximal nature of the test. A pilot study conducted by the present research group before the beginning of this investigation revealed that using the $13 \text{ km} \cdot \text{h}^{-1}$ threshold for the same sample assessed in this study would lead to elevated fatigue levels, thereby compromising the objectives of the test. In summary, the present results of the submaximal test, in conjunction with the physical performance test, suggest the effectiveness of the proposed periodization program.

The present results also suggest that during the in-season, compared with the preseason phase, the specific basketball training sessions presented higher intensity and lower volume. The eTL associated with accelerations and decelerations (mechanical load) increased during in-season, despite a reduction in training volume. This is a novel finding, demonstrating empirically that during the competition phase the coaches prescribed training content, based mainly on high-intensity actions, to mimic real competition demands. During intermittent team sports, players perform numerous accelerations and decelerations both with and without changes of direction (5). Additionally, these types of movements are likely to elicit substantial physical demands

and associated physiological responses (35). Indeed, given that basketball-specific actions involve whole-body displacement in forward, backward, lateral, and vertical directions, the accelerometer-derived measures are a useful approach to monitor eTL in basketball players (42), and perhaps a more appropriate method to monitor the eTL undertaken by professional basketball players. Additionally, the present findings not only reveal the usefulness of monitoring training by means of mechanical load but also these trends could improve our understanding of how professional basketball players train, with special reference to the technical and tactical sessions. This could provide valuable information to coaches on optimizing training during both the preseason and in-season periods.

An important but expected finding of the present investigation concerning monitoring of iTL using session-RPE and TRIMP is that, these were affected by “training volume.” These findings corroborate previous studies in team-sport athletes that demonstrated the influence of training volume in iTL. For example, Nunes et al. (41) showed that iTL responses of elite female basketball players were aligned with the preprogrammed overloading (higher training volume) and tapering phases (lower training volume). Indeed, Jeong et al. (24) reported significant higher iTL in the preseason than in-season, which reflected the higher volume of training performed during preseason. Additionally, the present results suggest that the increase in training session intensity, mainly due to the increase in actions requiring changes in direction, accelerations and decelerations, high-speed sprints, and other related specific basketball actions might lead to a higher ML.

The similar responses among iTL measures suggest that these markers represent a similar construct. However, these relationships are shown to be affected by different types of training sessions in team sports (11,30). Despite these discrepancies between session-RPE and HR, the results of this study add to the literature by demonstrating that both iTL and TRIMP markers were not only sensitive to the volume of the training (a relevant marker related to eTL) but also showed higher preseason values compared with the in-season phase in professional basketball players. It is important to report that the higher iTL and TRIMP values during the preseason phase were expected. (43). It was also expected that the intensity of the technical and tactical sessions would increase during the in-season phase as a reflection of training periodization. This hypothesis was based on the assumption that for team sports training periodization, a greater training volume would be completed during the preseason, as demonstrated previously (24,36) and proposed by others (43). Even considering that to date, there are still no comparative studies examining the efficacy of this approach (36), this training content distribution is widely accepted and used by coaches (8).

A major limitation of training intervention studies generally is the lack of a control group (46), and the reader should

be aware of this when interpreting the present results. Despite this limitation, the present data confirmed the initial hypothesis that the adopted training program would enhance physical test performances. Additionally, another potential limitation of this study is the sample size, especially the number of players per position; however, it should be highlighted that because of the difficulty in assessing team sports athletes and the number of athletes belonging to these teams, studies in professional team athletes are usually conducted with small sample sizes. For example, Manzi et al. (31) investigated the training load of 8 professional Italian basketball players across 3 different training weeks. Moreira et al. (39) examined the muscle soreness, blood muscle damage markers, muscle strength, and agility following an official basketball match in 11 professional women players. Montgomery et al. (35) investigated 11 basketball players to characterize the physical and physiological responses of different basketball practice drill and matches. Even recognizing the limitations of using a small sample size in this study, the results presented here add important empirical observations to the literature, in particular, to work related to training periodization, training load, and physical responses in elite team sport athletes.

In summary, the present results suggest that the adopted periodization, characterized by a higher preseason training volume and both higher intensity and increased specific basketball actions during the in-season led to a significant increase in physical performance in professional basketball players. Through analysis of the dynamics of iTL and eTL measures, the results suggest that the “volume” might mainly influence the iTL and, on the other hand, the intensity might considerably affect the eTL measures. Moreover, it can be inferred that fitness status indicators such as HRE and HRR should be used in conjunction with the physical performance test to evaluate the effectiveness of the proposed periodization program.

PRACTICAL APPLICATIONS

The present results indicate that coaches should combine iTL and eTL monitoring approaches to enable more efficient means of monitoring training load in basketball. Notably, basketball coaches should use the session-RPE method combined with accelerometer-derived measures to assess iTL and eTL, respectively, for both preseason and in-season phases. Additionally, coaches should be aware that using the ML, which is an accelerometer-derived measure, would be a very useful approach and even a more appropriate method to monitor the eTL undertaken by professional basketball players. Finally, the present findings suggest that because of the congruence with the improvement in physical performance, and a relevant change in fitness status indicators (HRE and HRR), it appears reasonable to propose these measures to monitor fitness status of professional basketball players. The submaximal nature of the running test is an advantage for use in professional

basketball practical settings especially during the in-season phase, when practitioners do not want players to perform maximal and exhaustive tests.

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